CPE412 Pattern Recognition

Week 2

Feature Formation and Feature Selection

Dr. Nehad Ramaha,
Computer Engineering Department
Karabük Universities

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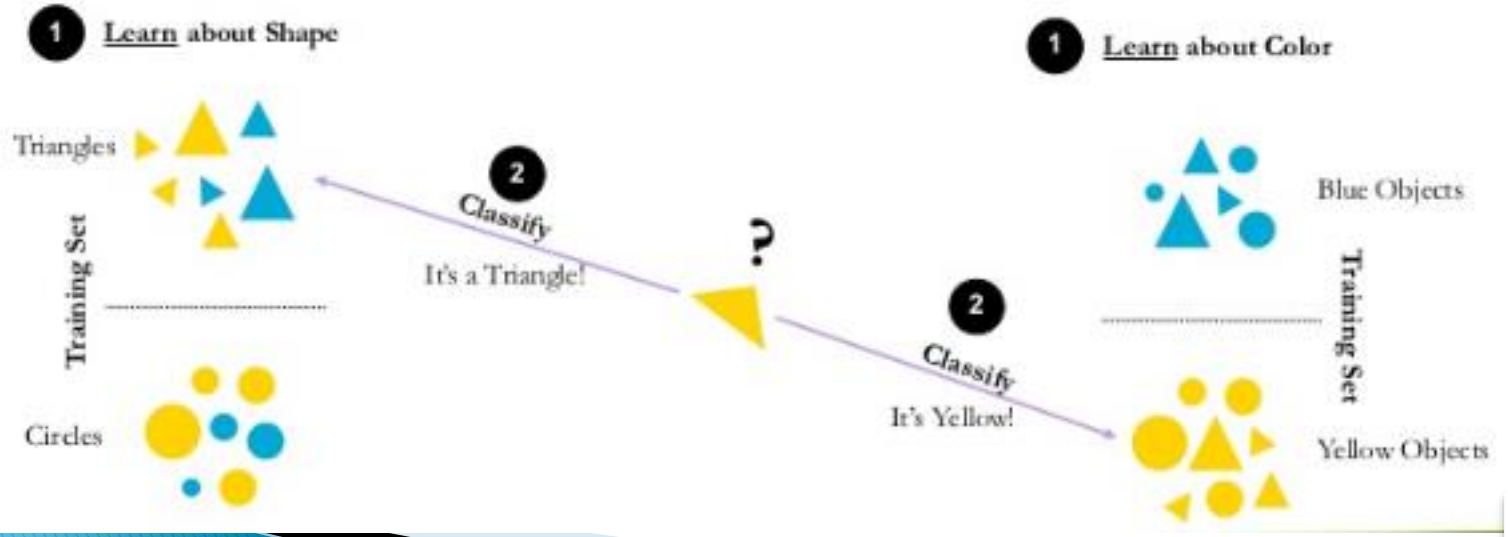
Pattern Recognition

- How can a machine learn the rule from data?
- Supervised learning: a teacher provides a category label or cost for each pattern in the training set.
  - Classification
- Unsupervised learning: the system forms clusters or natural groupings of the input patterns (based on some similarity criteria).
  - Clustering
- Reinforcement learning: no desired category is given but the teacher provides feedback to the system such as the decision is right or wrong.

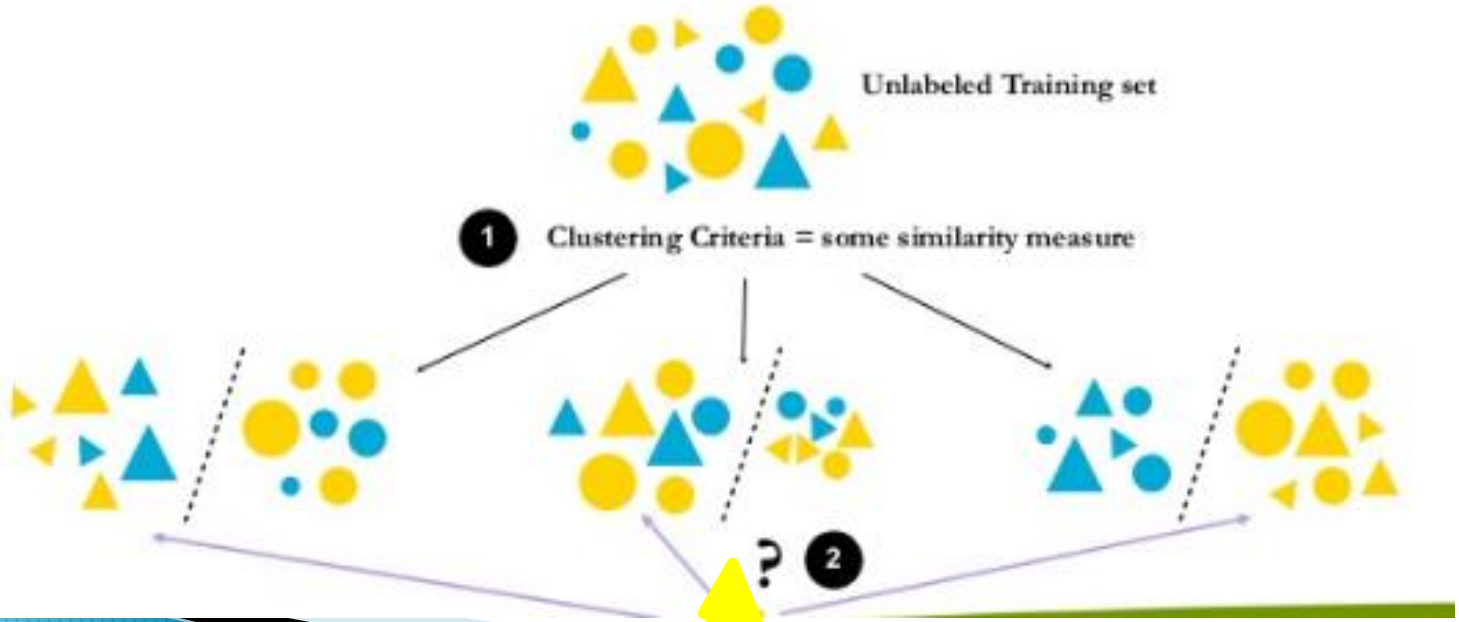
Pattern Recognition

- **Supervised Training/Learning**
  - a "teacher" provides labeled training sets, used to train a classifier



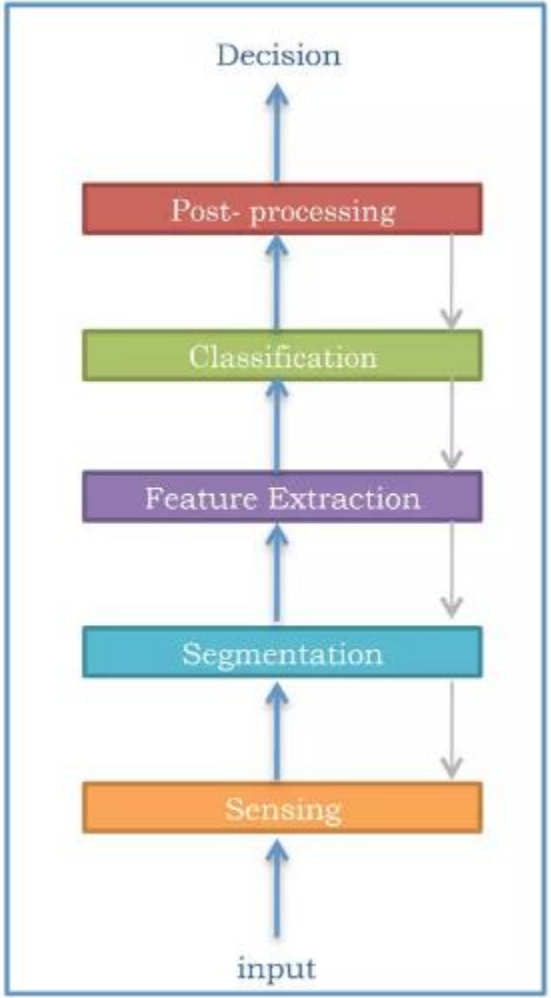
Pattern Recognition

- **Unsupervised Training/Learning**
  - No labeled training sets are provided
  - System applies a specified clustering/grouping criteria to unlabeled dataset
  - Clusters/groups together "most similar" objects (according to given criteria)



Pattern Recognition Process

- Data acquisition and sensing:
  - Measurements of physical variables.
  - Important issues: bandwidth, resolution , etc.
- Pre-processing:
  - Removal of noise in data.
  - Isolation of patterns of interest from the background.
- Feature extraction:
  - Finding a new representation in terms of features.
- Classification
  - Using features and learned models to assign a pattern to a category.
- Post-processing
  - Evaluation of confidence in decisions.



Features

- Features are properties of an object:
  - Ideally representative of a specific type (i.e. class) of objects
  - Compact (memory efficient)
  - Computationally simple (CPU efficient)
  - Perceptual relevant (if trying to implement a human inspired classifier)

Features

- Take a group of graphical objects
  - Possible features:
    - Shape
    - Color
    - Size
    - ...
  - Allows to group them into different classes:

SHAPE

SIZE

COLOR

Feature Vectors

- Usually a single object can be represented using several features, e.g.
  - **x1** = shape (e.g. nr of sides)
  - **x2** = size (e.g. some numeric value)
  - **x3** = color (e.g. rgb values)
  - ...
  - **xd** = some other (numeric) feature.

- **X** becomes a feature vector
  - x is a point in a d-dimensional **feature space**.

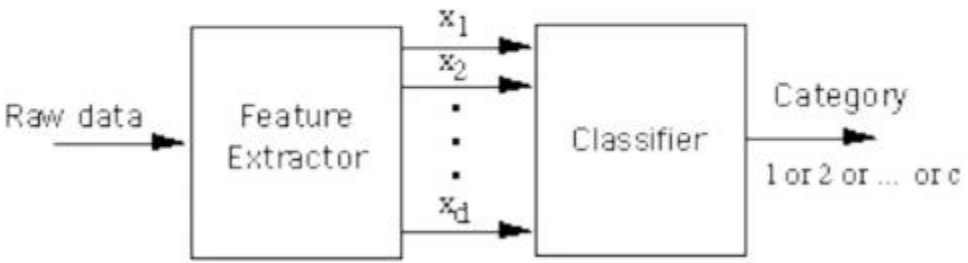
$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_d \end{bmatrix}$$

Feature Vectors

- Example of a 2D Feature Space
  - **x1** = shape (e.g. nr of sides)
  - **x2** = size (e.g. some numeric value)

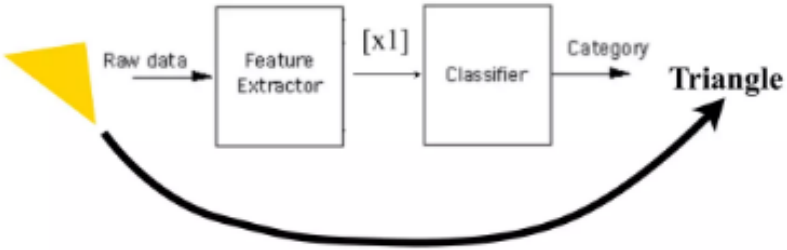
The Classical Model for PR

1. A **Feature Extractor** extracts features from raw data (e.g. audio, image, weather data, etc)
2. A **Classifier** receives X and assigns it to one of c categories, Class 1, Class 2, ..., Class c (i.e. labels the raw data).



The Classical Model for PR

- Example: classify graphic objects according to their shape
  - Feature extracted:
    - nr. of sides (x1)
  - Classifier:
    - 0 sides => circle
    - 3 sides => triangle
    - (4 sides => rectangle)



  - How does the classifier know that a circle has no sides and that a triangle has 3 sides?!

Example: Sorting Fish

Salmon

Sea Bass

- Problem: Sorting incoming fish on a conveyor belt according to species.
- Assume that we have only two kinds of fish:
  - sea bass,
  - salmon.

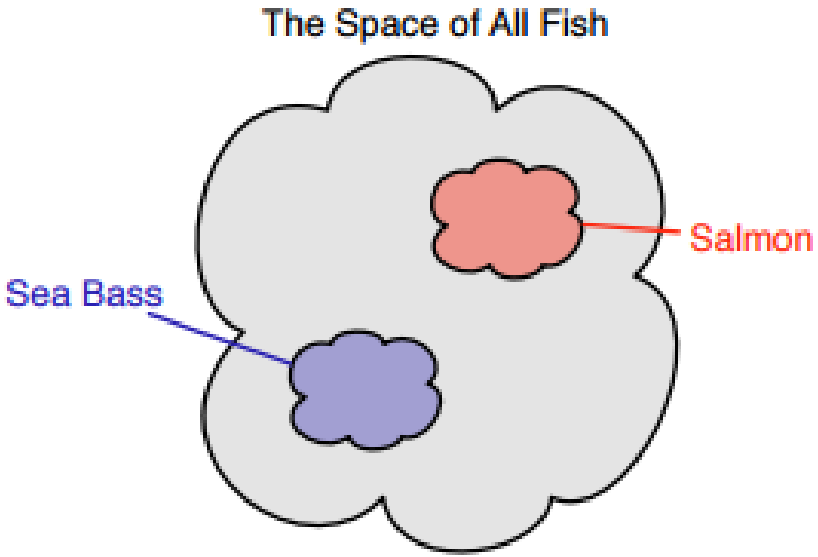
Figure 12: Picture taken from a camera.

Decision process

- What kind of information can distinguish one species from the other?
  - length, width, weight, number and shape of fins, tail shape, etc.
- What can cause problems during sensing?
  - lighting conditions, position of fish on the conveyor belt, camera noise, etc.
- What are the steps in the process?
  - capture image $\rightarrow$ isolate fish $\rightarrow$ take measurements $\rightarrow$ make decision

Patterns and Models

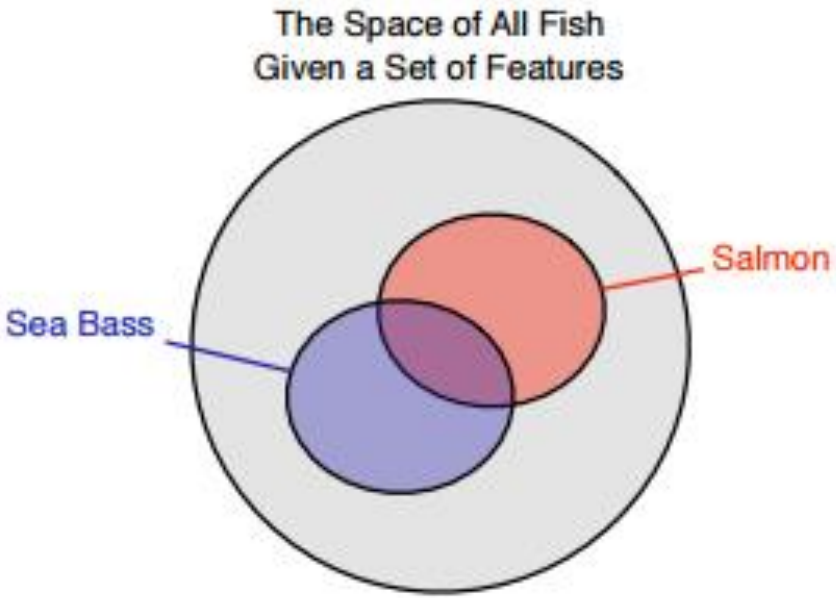
Ideal State Space



- Clear that the populations of salmon and sea bass are indeed distinct.
- The *space of all fish* is quite large. Each dimension is defined by some property of the fish, most of which we cannot even measure with the camera.

Patterns and Models

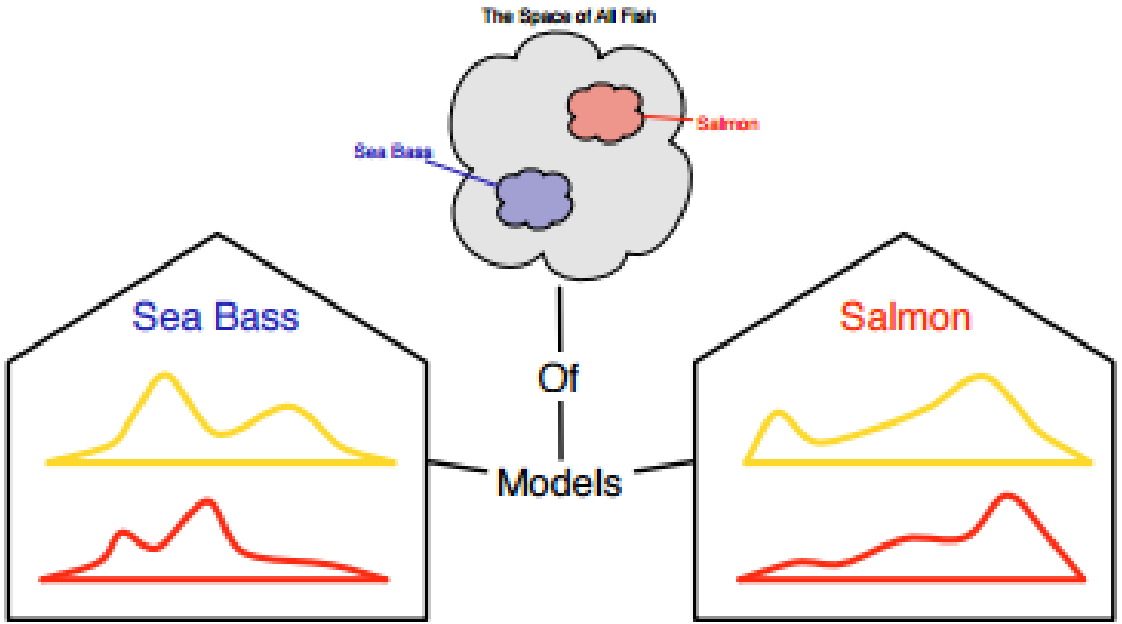
Real State Space



- When we choose a set of possible features, we are projecting this very high dimension space down into a lower dimension space.

Patterns and Models

Models



- We build a model of each phenomenon we want to classify, which is an approximate representation given the features we've selected.

Selecting features

- Assume a fisherman told us that a sea bass is generally longer than a salmon.
- We can use length as a *feature* and decide between sea bass and salmon according to a threshold on length.
- How can we choose this threshold?

Selecting features

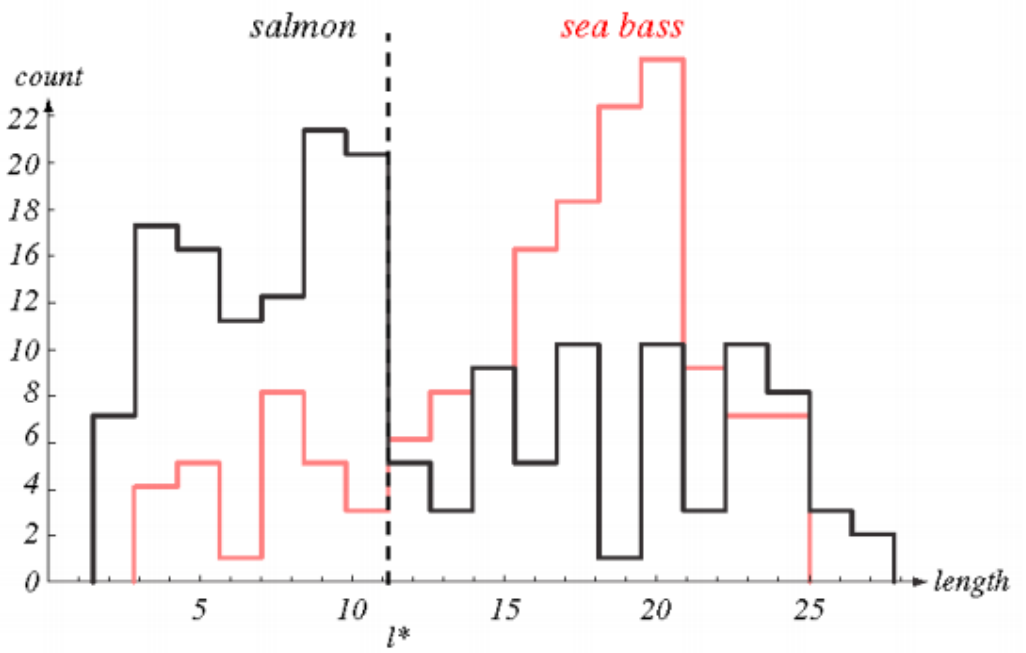


Figure 13: *Histograms* of the length feature for two types of fish in *training samples*. How can we choose the threshold l^* to make a reliable decision?

Selecting features

- Even though sea bass is longer than salmon on the average, there are many examples of fish where this observation does not hold.
- Try another feature: average lightness of the fish scales.

Selecting features

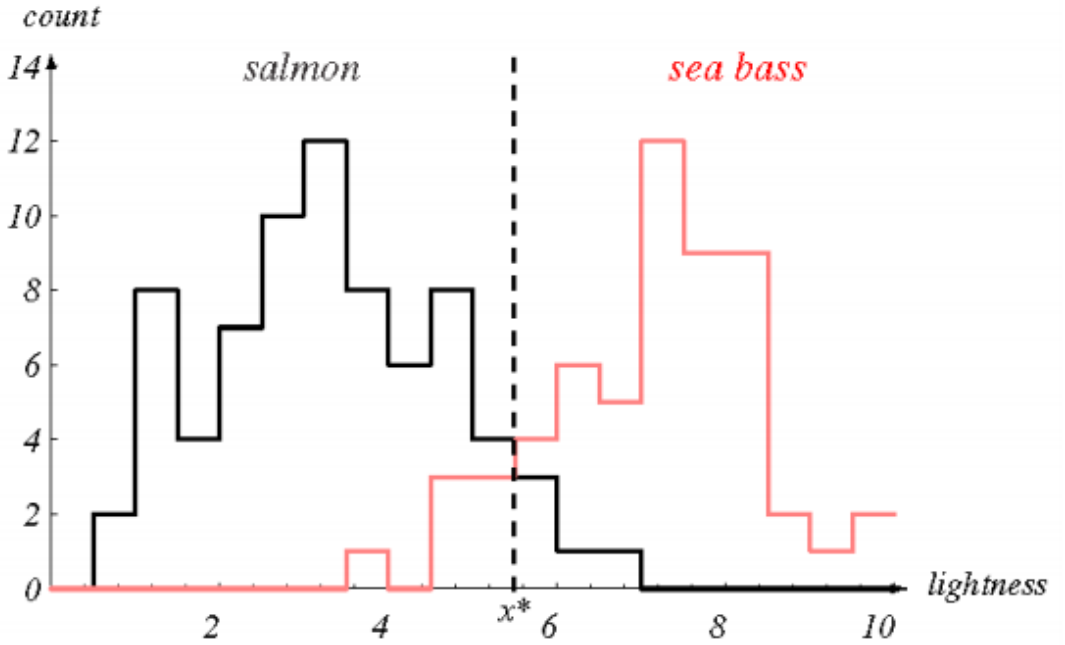


Figure 14: Histograms of the lightness feature for two types of fish in training samples. It looks easier to choose the threshold x^* but we still cannot make a perfect decision.

Cost of error

- We should also consider *costs of different errors* we make in our decisions.
- For example, if the fish packing company knows that:
  - Customers who buy salmon will object vigorously if they see sea bass in their cans.
  - Customers who buy sea bass will not be unhappy if they occasionally see some expensive salmon in their cans.
- How does this knowledge affect our decision?

Multiple features

- Assume we also observed that sea bass are typically wider than salmon.
- We can use two features in our decision:
  - lightness: x_1
  - width: x_2
- Each fish image is now represented as a point (*feature vector*)

$$\mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$$

in a two-dimensional *feature space*.

Multiple features

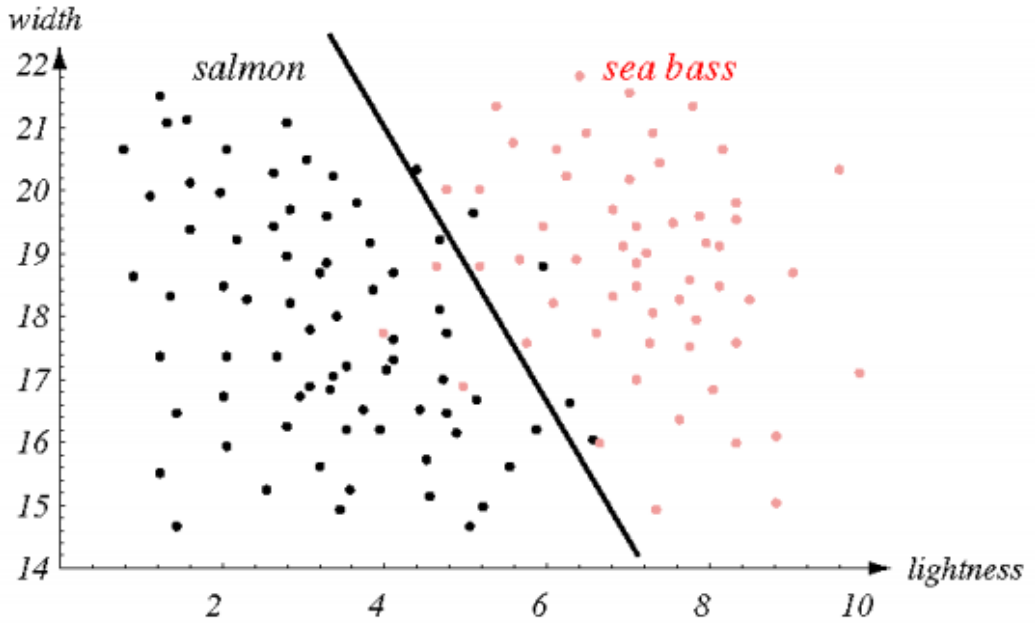


Figure 15: *Scatter plot* of lightness and width features for training samples. We can draw a *decision boundary* to divide the feature space into two regions. Does it look better than using only lightness?

Multiple features

- Does adding more features always improve the results?
  - Avoid unreliable features.
  - Be careful about correlations with existing features.
  - Be careful about measurement costs.
  - Be careful about noise in the measurements.
- Is there some *curse* for working in very high dimensions?

Decision boundaries

- Can we do better with another decision rule?
- More complex models result in more complex boundaries.

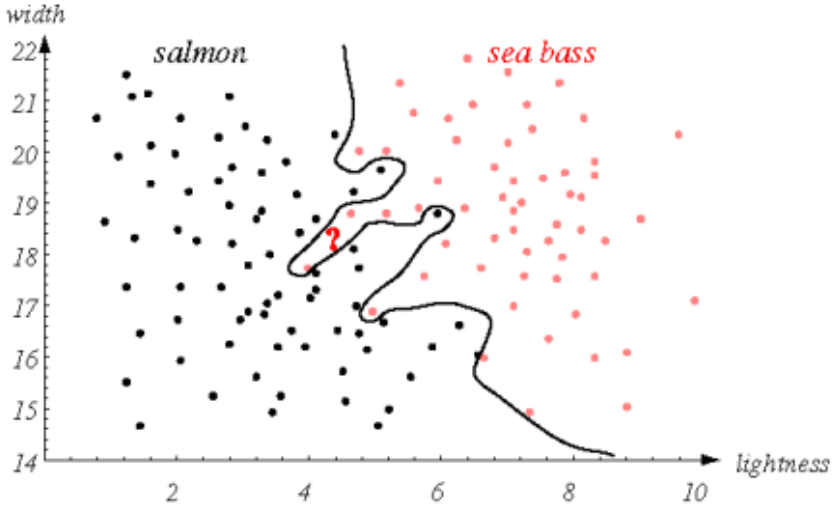


Figure 16: We may distinguish training samples perfectly but how can we predict how well we can *generalize* to unknown samples?

Decision boundaries

- How can we manage the *tradeoff* between complexity of decision rules and their performance to unknown samples?

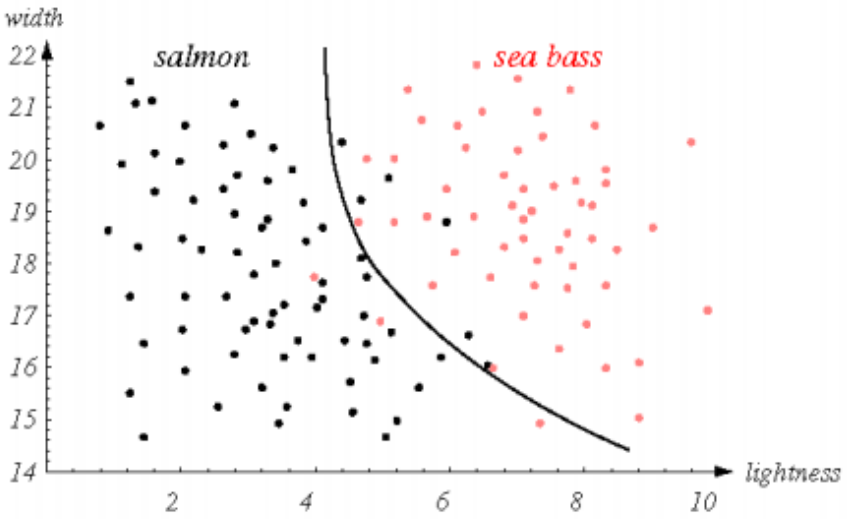


Figure 17: Different criteria lead to different decision boundaries.

Thanks ☺